

1. Consider the following relational database:

classroom(building, room_number, capacity)
department(dept_name, building, budget)
section(course_id, sec_id, semester, year, building)
course(course_id, title, dept_name, credits)
instructor(ID, name, dept_name, salary)
teaches(ID, course_id, sec_id, semester, year)

here, primary keys are underlined. Give an expression in the SQL to express each of the following queries:

- i. Find the department names of all instructors.
- ii. Retrieve the names of all instructors, along with their department names and department building name.
- iii. Find the names of all departments whose building name includes the substring 'Watson'.
- iv. Show the list of entire instructor in descending order of salary.
- v. Find the names of instructors with salary amounts between \$90,000 and \$100,000.
- vi. Find the set of all courses taught either in Fall 2017 or in Spring 2018, or both.
- vii. Find the average salary of instructors in the Computer Science department.
- viii. Find the average salary in each department.

Find the average salary of instructors in those departments where the average salary is more than \$42,000.

2. Consider the following relational database:

classroom(building, room_number, capacity)
student(ID, name, dept_name, tot_cred)
department(dept_name, building, budget)
takes(ID, course_id, sec_id, semester, year, grade)
section(course_id, sec_id, semester, year, building)
course(course_id, title, dept_name, credits)
instructor(ID, name, dept_name, salary)
teaches(ID, course_id, sec_id, semester, year)

here, primary keys are underlined. Give an expression in the SQL to express each of the following queries:

- i. Retrieve the number of instructors in each department.
- ii. Find the names of all departments whose building name beginning with 'Waz'.
- iii. Find those departments where the average salary of the instructors is more than \$42,000.
- iv. Show the list of entire instructor in ascending order of salary.
- v. Delete all instructors with a salary between \$13,000 and \$15,000.
- vi. List the names of students along with the titles of courses that they have taken.
- vii. Find all students who have not taken a course.

Find all Physics courses offered in the Fall 2017 semester in the Watson building by defined a view *physics_fall_2017*.

3. Consider the following employee database
employee(employee_name, street, city)
works(employee_name, company_name, salary)
company(company_name, city)
manages(employee_name, manager_name)

Figure: A-1

Consider the employee database of Figure A-1, where the primary keys are underlined. Give an expression in SQL for each of the following queries.

- i) Find the company that has the most employees.
- ii) Find those companies whose companies earn a higher salary, on average, than the average salary at “First Bank Corporation”.

4. Consider the following employee database
employee(employee_name, street, city)
works(employee_name, company_name, salary)
company(company_name, city)
manages(employee_name, manager_name)

Figure: A-1

Give an expression in the relational algebra to express each of the following queries:

- i) Find the names of all employees who work for “First Bank Corporation”.
- ii) Find the names and cities of residence of all employees who work for “First Bank Corporation”.
- iii) Find the names, street address and cities of residence of all employees who work for “First Bank Corporation” and earn more than BDT 10,000.

5. Consider the following relational database:

department(dept_name, building, budget)
section(course_id, sec_id, semester, year, building)
course(course_id, title, dept_name, credits)
instructor(ID, name, dept_name, salary)
teaches(ID, course_id, sec_id, semester, year)

Give an expression in the relational algebra to express each of the following queries:

- i. Find all instructors with salary greater than \$90,000.
- ii. Find the set of all courses taught in both the Fall 2017 and the Spring 2018 semesters.
- iii. Find the ID and name of those instructors who earn more than the instructor whose ID is 12121.
- iv. Find the name of an instructor who has taught course.

Find the names of all instructors except the Physics department.

6. Consider the following employee database:

employee (person_name, street, city)

works (person_name, company_name, salary)

company (company_name, city)

Give an expression in the relational algebra to express each of the following queries:

- i. Find the name of each employee who lives in city "Pabna".
- ii. Find the name of each employee who lives in "Pabna" and whose salary is greater than \$100,000.
- iii. Find the name of each employee in square company with a salary between \$90,000 and \$100,000.
- iv. Show the name of all employees who lives in city either "Pabna" or "Natore".

Find the name of each employee who earns more than the employee whose name is Rahim.

7. Consider the following employee database where primary keys are underlined. Give an SQL DDL definition of this database then write SQL expression for each of the queries given bellow.

employee (employee_name, street, city)

works (employee_name, company_name, salary)

company (company_name, city)

manages (employee_name, manager_name)

Fig: Employee database.

- i. Find the names and cities of all employees who work in abc bank ltd.

Find the names, street address and cities of all employees who work in abc bank ltd and earn more than BDT 50000.